



Statistics of Financial Markets

Chapter 13: Systemic Risk



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Outline

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CoVaR

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References

Motivation

- ▣ Lessons from the 2007–2009 Global Financial Crisis
- ▣ Why individual risk measures are insufficient for systemic events
- ▣ Macroprudential regulation and the need for systemic risk quantification

Systemic Risk Measures

- Overview of systemic risk measurement approaches
- Marginal Expected Shortfall (MES)
- Systemic Expected Shortfall (SES)
- SRISK: systemic risk contribution based on capital shortfall
- Comparison of different measures

CoVaR

- ▣ Adrian and Brunnermeier's (2016) Conditional Value at Risk
- ▣ $\text{CoVaR}_q^{j|i}$: VaR of institution j conditional on institution i being in distress
- ▣ ΔCoVaR : contribution to systemic risk
- ▣ Estimation via quantile regression
- ▣ Time-varying CoVaR and state variables

Financial Risk Meter (FRM)

- ▣ LASSO-based approach to measuring tail risk interconnectedness
- ▣ Penalized quantile regression across financial institutions
- ▣ FRM index construction from estimated network density
- ▣ Real-time monitoring of systemic risk
- ▣ Applications to different financial systems

Network Analysis

- ▣ Financial networks and contagion channels
- ▣ Adjacency matrices from Granger causality or tail dependence
- ▣ Network topology metrics: degree, centrality, clustering
- ▣ Visualization and interpretation of financial networks

Applications

- ▣ Computing CoVaR for major financial institutions
- ▣ FRM analysis during crisis periods
- ▣ Network visualization of systemic risk
- ▣ Python implementation

 `SFM_ch13_systemic_risk`

Conclusions

- ▣ Summary of systemic risk measurement methodologies
- ▣ Strengths and limitations of each approach
- ▣ Policy implications and regulatory applications

References

- Adrian, T., Brunnermeier, M.K. (2016). CoVaR.
- Acharya, V.V., Pedersen, L.H., Philippon, T., Richardson, M. (2017). Measuring systemic risk.
- Härdle, W.K., Wang, W., Yu, L. (2016). TENET: Tail-Event driven NETwork risk.